Practical no. 11

Aim - Selenium Server (Grid) Configuration and Remote Execution

Steps 1 – Install Java, Maven
Prompt: *java -version*
javac -version
Check Maven: *mvn -version*
You should see: Apache Maven 3.x.x

Step 2 - Open Environment Variables. Open system variables and add following all path -
C:\Program Files (x86)\Java\jdk1.8.0_144\bin
C:\Program Files (x86)\Java\jre1.8.0_144\bin
C:\Program Files\Java\jdk-21.0.12\bin
C:\Program Files\Java\jre1.8.0_461\bin
D:\STQA\apache-maven-3.9.16\bin

Step 3- Download Selenium Server
C:\selenium-grid\selenium-server-4.35.0.jar

Step 4 - Start the selenium hub –
Open cmd(1)
C:\Users\ITCS> *cd C:\selenium-grid*
C:\selenium-grid>java -jar selenium-server-4.35.0.jar hub
Find the Hub IP Address
Open a browser on the Hub machine.
Go to: http://localhost:4444 eg- **http://192.168.105.107:4444** You should see the Selenium Grid UI.
Verify the Grid Status –
http://192.168.105.107:4444/status

Step 5 - Configure Node 1 — Chrome
Start Chrome Node. On the Node machine: Open cmd(2) -
cd C:\selenium-grid
Then run: *java -jar selenium-server-4.35.0.jar node --hub http://192.168.105.107:4444*
The Node will register itself with the Hub.
Step 6 - Configure Node 2 — Firefox
Step 7 - Check Network Connectivity
Open cmd(3) -
cd C:\selenium-grid
ping 192.168.105.107 //ping localhost

Open Windows PowerShell, also check port 4444 -
Test-NetConnection 192.168.105.107 -Port 4444

You want: TcpTestSucceeded : True

Step 8 - Create Your Java Maven Project

Create a Maven project. Example structure:

Open cmd(4) –

cd..

C:\>mkdir selenium-grid-demo

C:\>cd selenium-grid-demo

>mkdir src

>mkdir src\main

>mkdir src\main\java

C:\selenium-grid-demo>notepad src\main\java\CrossBrowserGridTest.java

```
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeOptions;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.remote.RemoteWebDriver;
import java.net.URL;
public class CrossBrowserGridTest {
private static final String GRID_URL =
"http://192.168.105.107:4444";
private static final String TEST_URL =
"https://www.google.com";
public static void main(String[] args) throws Exception {
runChromeTest();
runFirefoxTest();
}
public static void runChromeTest() throws Exception {
WebDriver driver = null;
try {
ChromeOptions options = new ChromeOptions();
driver = new RemoteWebDriver(
new URL(GRID_URL),
options
);
driver.get(TEST_URL);
String title = driver.getTitle();
System.out.println("Browser: Chrome");
System.out.println("URL: " + driver.getCurrentUrl());
System.out.println("Title: " + title);
if (title.contains("Google")) {
System.out.println("Chrome: PASS");
} else {
System.out.println("Chrome: FAIL");
}
} finally {
if (driver != null) {
driver.quit();
}
}
}
public static void runFirefoxTest() throws Exception {
WebDriver driver = null;
try {
FirefoxOptions options = new FirefoxOptions();
driver = new RemoteWebDriver(

new URL(GRID_URL),
options
);
driver.get(TEST_URL);
String title = driver.getTitle();
System.out.println("Browser: Firefox");
System.out.println("URL: " + driver.getCurrentUrl());
System.out.println("Title: " + title);
if (title.contains("Google")) {
System.out.println("Firefox: PASS");
} else {
System.out.println("Firefox: FAIL");
}
} finally {
if (driver != null) {
driver.quit();
}
}
}
}
```

Step 9 – Create pom.xml notepad file.

C:\selenium-grid-demo>notepad pom.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
 xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0  https://maven.apache.org/xsd/maven-4.0.0.xsd">
 <modelVersion>4.0.0</modelVersion>
 <groupId>com.selenium.grid</groupId>
 <artifactId>selenium-grid-demo</artifactId>
 <version>1.0-SNAPSHOT</version>
 <properties>
 <maven.compiler.source>17</maven.compiler.source>  <maven.compiler.target>17</maven.compiler.target>
<project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>  </properties>
 <dependencies>
 <dependency>
 <groupId>org.seleniumhq.selenium</groupId>
 <artifactId>selenium-java</artifactId>
 <version>4.35.0</version>
 </dependency>
</dependencies>
 <build>
 <plugins>
 <plugin>
 <groupId>org.codehaus.mojo</groupId>
 <artifactId>exec-maven-plugin</artifactId>  <version>3.5.0</version>
 <configuration>
 <mainClass>CrossBrowserGridTest</mainClass>  </configuration>
 </plugin>
 </plugins>
 </build>
</project>
```

Step 11 - Run the Test

Open cmd(4)

C:\selenium-grid-demo>mvn compile

C:\selenium-grid-demo>mvn exec:java

\